

CTEC1704 – LAB 4

Use the main text books from the resource list and other resources to help you answer the following questions.

COMPUTER SYSTEM

FOUNDATION

1. What is hardware?
2. What is clock speed measured in?
3. List the components of the central processing unit (CPU)
4. Which central processing unit (CPU) component makes logical decisions?
5. What is a register?
6. What is a core?
7. What are the three types of bus?
8. Which of these statements about Von Neumann architecture is true?
 - Only data is stored in main memory
 - Only instructions are stored in main memory
 - Data and instructions are both stored in main memory
9. Which register holds the address of the next instruction to be fetched from memory?

COMPUTER ARCHITECTURE

1. List 8 Great Ideas in Computer Architecture and explain each one.
2. What is the purpose of the CPU?
3. What does the control unit do?
4. What is the immediate access store for?
5. What does the arithmetic and logic unit do?
6. What can often be referred to as the registers in a CPU?
7. How many instructions can a CPU process at a time?
8. What is it called when the CPU carries out the action of an instruction?
9. What is clock speed measured in?
10. How many cycles per second would a 3 GHz processor do?